Stock Market Anomaly Detection – Capstone Report

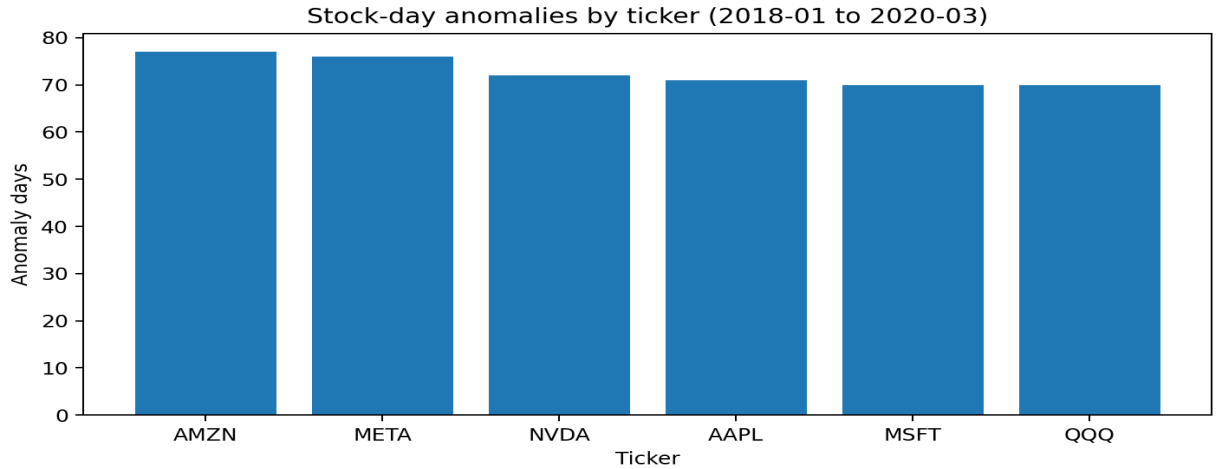
Goal: Detect unusual market days and stock days using only daily price and volume data, using leakage-safe rolling features and simple rules.

Component	Definition / Value
Universe	QQQ, AAPL, MSFT, NVDA, AMZN, META
Windows	W_return=63, W_volume=21, W_range=63 (past-only)
Stock-day anomaly rule	|ret_z|>2.5 OR volz>2.5 OR range_pct>95
Market-day anomaly rule	|market_ret| > rolling 95th percentile OR breadth < 0.3
Splits	Train=2018, Val=2019, Test=2020 Q1 (Jan–Mar)

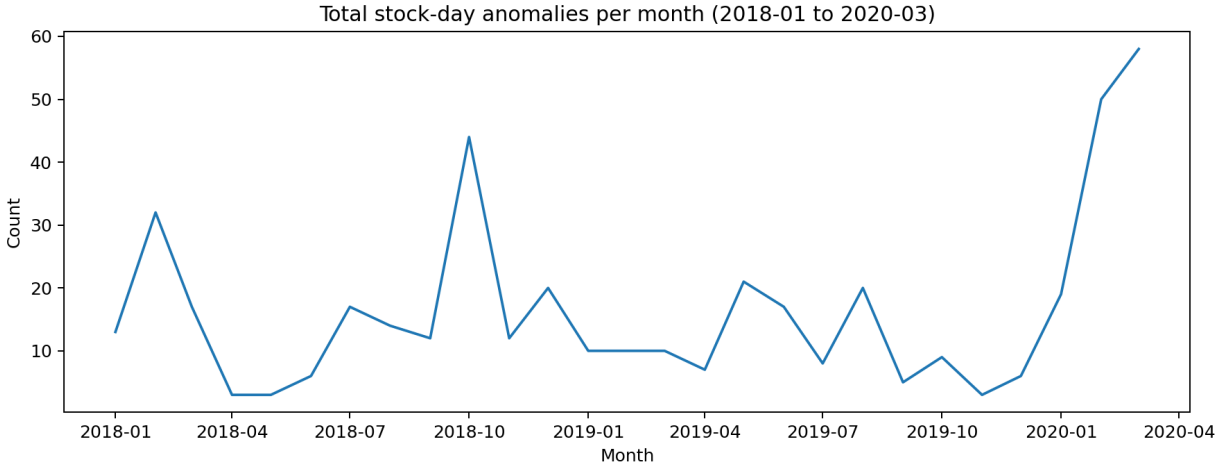
Metric	Value (2018-01 to 2020-03)
Stock-day anomalies flagged	436
Market days	565
Market-day anomalies flagged	192
Overlap (market-day flag AND any stock anomaly)	102

Figures

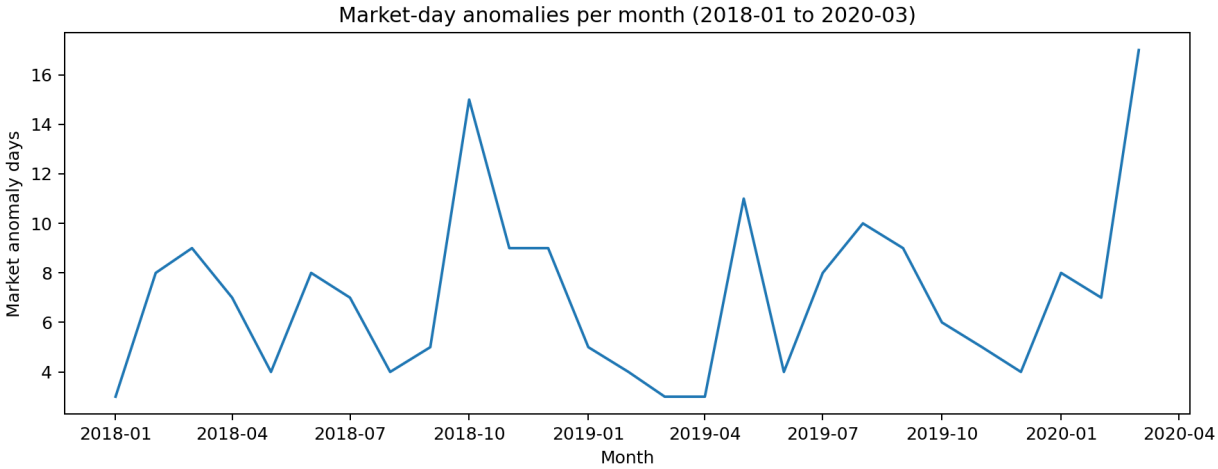
Stock-day anomalies by ticker



Total stock-day anomalies per month



Market-day anomalies per month


Market-day anomalies per month (2018-01 to 2020-03)
Market anomaly days
16
14
12
10
8
6
4
2018-01
2018-04
2018-07
2018-10
2019-01
2019-04
2019-07
2019-10
2020-01
2020-04
Month

Top 10 severe flagged stock-days (examples)

date	ticker	type	ret_z	volz	range_pct	why
2018-07-26	META	crash + volume_shock	-12.091	6.525	100.0	|ret_z| > 2.5; volz > 2.5; range_pct > 9
2019-06-03	META	crash + volume_shock	-4.678	6.374	100.0	|ret_z| > 2.5; volz > 2.5; range_pct > 9
2018-02-05	QQQ	crash + volume_shock	-5.765	3.922	100.0	|ret_z| > 2.5; volz > 2.5; range_pct > 9
2018-10-10	QQQ	crash + volume_shock	-5.782	3.401	100.0	|ret_z| > 2.5; volz > 2.5; range_pct > 9
2018-02-05	NVDA	crash + volume_shock	-4.073	3.793	100.0	|ret_z| > 2.5; volz > 2.5; range_pct > 9
2018-02-02	AAPL	crash + volume_shock	-4.237	3.461	100.0	|ret_z| > 2.5; volz > 2.5; range_pct > 9
2018-10-10	AMZN	crash + volume_shock	-4.616	3.292	100.0	|ret_z| > 2.5; volz > 2.5; range_pct > 9
2018-10-10	MSFT	crash + volume_shock	-5.113	3.114	100.0	|ret_z| > 2.5; volz > 2.5; range_pct > 9
2019-06-03	AMZN	crash + volume_shock	-3.472	3.367	100.0	|ret_z| > 2.5; volz > 2.5; range_pct > 9
2018-03-27	NVDA	crash + volume_shock	-3.092	3.683	100.0	|ret_z| > 2.5; volz > 2.5; range_pct > 9